

Computational classification of SOD2 protein in coral and symbiotic algae from public genomic and transcriptomic data

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ABSTRACT

SOD2, or Manganese/Iron Superoxide Dismutase, plays a crucial role in maintaining cellular health across the tree of life. Existing literature is vague in identifying types of SOD2 proteins expressed in both corals and their symbiotic algae; most agree on the presence of a) mitochondrial targeting MnSOD in corals, and b) likely Mn dependent SOD2 in algae. This paper presents evidence for four classes of SOD2 proteins between coral hosts and their symbiotic algae. Public genomic and transcriptomic data reveal mitochondrial targeting and secreted versions of SOD2 among a wide range of cnidaria species, and chloroplast targeting (with a bipartite leader architecture) and secreted versions of SOD2 among several symbiodinium species. The mature/catalytic region of SOD2 is well conserved within each class, but highly differentiated between classes. This paper presents the mature region HMM profiles for each class, with threshold scores high enough such that they are predictive of leader type.

Introduction

SOD2 proteins protect cells against oxidative damage by converting toxic superoxide radicals into hydrogen peroxide and diatomic oxygen. SOD2 proteins can bind either manganese (MnSOD) or iron (FeSOD) as co-factors; their structure consists of conserved residues His26, His74, Asp159, and His163 (human sequence coordinates) forming the anchors of a trigonal bipyramidal metal pocket and variations in the outer shell, based on bound metal, to ensure superoxide dismutase activity [1]. SOD2 is present throughout tree of life, with bacterial SOD2 often preferring Fe as cofactor, mammalian SOD2 almost always requires Mn as a cofactor to function in mitochondria, and metal preference not always phylogenetically constrained [2].

Existing literature is vague on the types of SOD2 proteins in coral hosts and their symbiotic algae. While there is consensus on the presence of mitochondrial MnSOD in corals, other forms of SOD2 in coral and symbiotic algae are rarely discussed/analyzed separately. [3] reports experimental evidence for secreted superoxide dismutase activity in *Stylophora pistillata* without identifying the source of such activity. [4] experimentally identifies two forms of MnSOD in *Hydra vulgaris*, a non-coral species in the cnidaria phylum. [5] computationally compares various annotated MnSOD and FeSOD sequences from various symbiodinium species, with grouping of sequences, based on putative annotation, inconsistent with findings of this paper. Indeed, most transcriptomic analyses in the field depend on blasting against UniProt or NCBI, or matching against the HMM profile K04564, a pan-metagenomic KEGG ortholog ([6]) for SOD2. Neither approaches provide resolution beyond putative superoxide dismutase activity.

This paper presents results from an unbiased computational search and grouping of SOD2 protein sequences, using NCBI deposited genome sequences from 36 coral and 10 symbiodinium species, as well as RNAseq datasets from 5 papers. The goal of this work is to present specific and unambiguous sequence-level profiles for SOD2 proteins in *Anthozoa* and *Dinophyceae*. Although these protein groups have features that are suggestive of biological and biochemical mechanism, the scope of this paper is limited to computational;

there is no experimental work, beyond referenced RNAseq experiments, to functionally validate the different classes of proteins, determine their metal preference, or likely cellular location.

Specifically, a protein sequence database was constructed from a) protein sequences submitted to and associated with NCBI genome accessions, b) protein sequences exhaustively detected using 25k+ KO HMM profiles ([6]) on 6-frame translations of genomic sequences submitted to and associated with NCBI genome accessions, and c) protein sequences generated from transcriptomes from 5 published RNAseq studies. 210 seed sequences meeting the K04564 HMM profile minimum threshold ([7]) were selected from this sequence database.

Seed sequences were further filtered to those containing PF00081 and PF02777 Pfam FeSOD N and C-terminal domains, using Pfam's gathering score ([8]), and having four key metal binding pocket residues ([1]). A *mature protein sequence* between (inclusive) the 3rd AA of PF00081 and 100th AA of PF02777 was extracted for each remaining seed sequence.

A leader discovery algorithm was employed to look for, in predicted exons if protein sequence was obtained through a gene model (above (a)) or in upstream 3-frame translations (above (b)) or in the protein sequence (above (c)), possible start codons upstream of each mature protein sequence. This algorithm produced zero (if no suitable start codon was found) or more full length SOD2 sequences for each mature sequence. This leader discovery step typically found predicted start sites for coral hosts, but was instrumental in finding complete SOD2 sequences in symbiotic algae genomic and transcriptomic data; tandom transcription ([9]) and alternative splicing sites ([10]) make gene prediction in dinoflagellates difficult otherwise. More on all of these in the Methods section below.

MUSCLE multi-sequence alignments ([11]) was used to group mature protein sequences. DeepLoc 2.1 ([12]) was used to determine possible target cellular compartment and leader type of each full length SOD2 sequence.

Results

Manual inspection of DeepLoc 2.1 results over full length SOD2 sequences suggests the presence of three patterns: a) a mitochondrial transit peptide (mTP) between 15 to 35 AAs upstream of PF00081 with sequence location predicted to be mitochondria, b) a signal peptide (SP) between 15 to 40 AAs upstream of PF00081 with sequence location predicted to be extracellular, and c) a signal peptide between 50 and 70 AAs upstream of PF00081 with sequence location predicted to be the golgi apparatus. Specifically, the first and last predictions were almost exclusively for coral hosts and symbiodiniums, respectively, while the middle prediction includes sequences for both hosts and symbionts.

Additional inspection using DeepLoc, on the SP and golgi apparatus sequences, reveal that once the SP is cleaved, DeepLoc predicts the remaining sequence as plastid bound. This observation is consistent with general consensus of bipartite leader architecture for chloroplast targeting proteins in cells that have gone through secondary endosymbiosis [13].

Based on these observations, leader discovery algorithm was adjusted to specifically look for start sites in these ranges, and the full length SOD2 sequences were separated into four groups: mTP in coral hosts, SP in coral hosts, SP+golgi in symbiodinium, and SP in symbiodinium. Multi-sequence alignments of mature protein sequences within each bucket showed high degree of conservation. Overall, 147 full length SOD2 sequences exhibited these DeepLoc called features. Table 1 shows the breakdown.

Four classes of SOD2 proteins, termed *mt_cnidaria*, *sp_cnidaria*, *sp_ctp_algae*, and *sp_algae*, are evident from the candidate sequences. Figure 1 shows a breakdown of the number of proteins per class, by taxonomy. Transcriptomic data are grouped in the first set of columns on the left with NULL as phylum, order, and genus values.

Table 1. Breakdown of full length SOD2 sequences

Group	Taxonomy	DeepLoc Predictions		Start codon upstream of PF00081	Num of Sequences
		Leader	Location		
mt_cnidaria	Coral host	mTP	Mitochondria	15 to 35 AAs	67
sp_cnidaria	Coral host	SP	Extracellular	15 to 40 AAs	39
sp_ctp_algae	Symbiodinium	SP	Golgi apparatus	50 to 70 AAs	32
sp_algae	Symbiodinium	SP	Extracellular	15 to 35 AAs	9

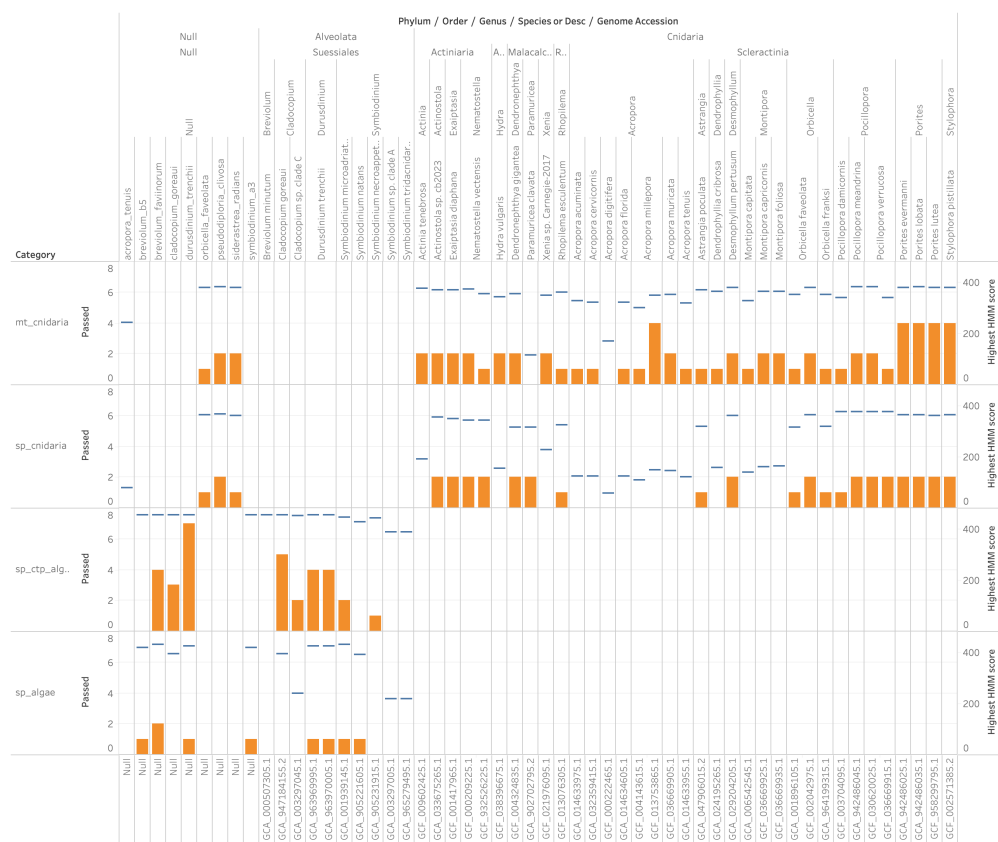


Figure 1. Sample figure caption.

Talk about transcriptomic data.

Numbers

147 was the used also to make figure 1

from the 147, 127 unique mature sequences were used to make HMM profiles for mature sequences

additional 37 reference sequeces, would have made 160, but we only used 155 for the tree generation, missing 5 sequences from PM34. this is figure 2.

Figure 2

IQtree of mature enzyme sequence (3 AA of PF00081 to 100 AA of PF02777) forms 4 clades, with sp_cnidaria being paraphyletic, the rest are monophyletic, hard to assign metals from this as sequence similarity may not correspond to metal preference. This establishes that



Figure 2. Sample figure caption.

A mitochondrial MnSOD exists across all corals

In coral hosts, the mitochondrial version of SOD2 includes of a mitochondrial transit peptide (mTP) XXX amino acids upstream of N-terminal PF00081 domain, and the secreted version includes a signal peptide (SP) XXX amino acids upstream of PF00081. These two classes are termed mt_cnidaria and sp_cnidaria, respectively.

High mature sequence similarity. HMM scores. Very likely Mn.

What about Mt? Likely different across species? Or also conserved?

Evidence of a secreted SOD2, with unknown metal preference, shown in the genome and transcriptome of some coral species

Paraphyletic, evolutionary lineage tree forms clusters by type: sea anemone, stony, soft.

Transcript counts from one RNAseq experiment

Missing from acropora genomes, which explains why a. tenuis transcripts does not have such a sequence.

Just one study. But uncertain if it's linked to this.

A likely chloroplast targeting SOD2 protein, with a bipartite leader, is conserved across multiple species of symbiodinium

In algae, the chloroplast targeting version of SOD2 includes a bipartite leader. DeepLoc 2.1 consistently finds a SP leader XXX amino acids upstream of PF00081, and also consistently categorizes the SP-cleaved copies of the proteins as localized to plastid. DeepLoc employs different mechanisms (or "heads" in their ML architecture) for classifying localization and finding leader sequences, and was not trained on symbiodiniums. The SP cleavage point sits XXX amino acids upstream of PF00081, suggesting a bipartite leader architecture of XXX AAs of SP, YYY AAs of cTP, then PF00081. This class is termed sp_ctp_algae, in figures and tables in this paper.

Separately, a secreted version of SOD2 with only a SP, XXX amino acids upstream of PF00081, also exist in algae. This class is termed sp_algae.

A fourth class of SOD2, with distinct mature protein sequence, may exist in symbiodinium, with weak expression evidence

May be secreted? Transcript counts from one RNAseq experiment suggest very weak if any expression.

The mature protein sequences of the four classes are cleanly distinguishable from each other and covers almost all of detectable K04564 sequences

The mature enzyme sequence, w/o leader, form distinctive clades that do not overlap. Furthermore the HMM threshold scores for mature enzyme group profiles show a large drop with F=1 (i.e. no FPs or FNs) at very clear boundaries. Mature enzyme sequence correlates with leader, and appear to also with target compartment

Generated 202 K04564 matches from our genomic and RNAseq datasets, reduced down to 114 unique mature enzyme sequences that possess all key residues and spans from 3rd AA of PF00081 to 100 AA of PF02777, so removing any partial matches due to bad genome assemblies or truncated gene models.

this is all_k04564_matches_mature_region.faa and the .tsv file

12 out of 144 fell below the above mature profile thresholds, rest belong to one of the 4 clades

4 are similar to or is CAE7459428.1 and is just below the threshold for sp_ctp_algae; CAE7459428.1 would have been a hit for sp_ctp_algae if its SP signal was stronger.

8/144, or 6

Final HMM profile for each group, with threshold scores for full protein and probably mature proteins, and rules

c. goreau should have had one, but in CAL1167723.1, PF00081 did not match well, if we define PF00081 from position 11 not position 3, may have been ok

Mature HMM vs leader

Should be SP-cPT? DeepLoc reports Golgi as targets for both but do not report SP for them, SignalP reports weak SP, but sequence kind of matches BGNK01003376.1_K04564_a3776c6c u60 OZ257276.1_K04564_ace37290 u60

Should be SP? These are the same sequence. DeepLoc thinks they are not SP and for plastid. Mature sequence definitely matches better for SP_algae but is below threshold. CAL1167723.1_with_leader_u21_PF00081_M evm.model.scf7180000356577.1_with_leader_u21_PF00081_M Issue is that we are missing 11 bps of PF00081

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